The Thermodynamics of Thermal Injury:

A Mathematical and Mechanistic Account of Heat-Induced Cardiac and Renal Damage

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Abstract

Heat exposure injures the heart and kidneys through distinct but coupled physical mechanisms. In the heart, the governing relation is an oxygen supply-demand mismatch: thermoregulatory increases in cardiac work raise myocardial oxygen demand while coronary perfusion pressure and diastolic filling time fall, producing ischemia when demand outpaces supply. In the kidney, the governing relation is a redistribution of cardiac output away from the renal circulation toward the cutaneous bed, compounded by hyperthermia-driven oxidative and cytotoxic injury. This paper formalizes both mechanisms, reviews the quantitative evidence for each, and models how a state actor could exploit these physiological vulnerabilities as a weapon. The weaponization pathway operates through environmental and infrastructural manipulation rather than novel technology: it requires only the deliberate removal of protective conditions and the selection of physiologically vulnerable populations, such that the resulting harm is individually indistinguishable from natural causes.

Keywords: heat stress, myocardial ischemia, acute kidney injury, thermoregulation, oxidative stress, weaponization, physiological vulnerability

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1 The Cardiac Mechanism: A Demand-Supply Inequality

Let $D(t)$ denote myocardial oxygen demand and $S(t)$ coronary oxygen supply at time t. Ischemia emerges when

$$D(t) > S(t). \tag{1}$$

Heat stress operates on both sides of this inequality. Demand rises because thermoregulation requires an increase in cardiac output Q and heart rate HR; supply falls because the same thermoregulatory redistribution that cools the skin reduces coronary perfusion pressure and shortens diastole, the interval during which coronary flow predominantly occurs.

1.1 Demand

Myocardial oxygen consumption scales approximately with the rate-pressure product:

$$D \propto HR \cdot SBP, \tag{2}$$

where SBP is systolic blood pressure. In a randomized trial of exercise-induced heat stress, the systolic pressure-time integral—an index of myocardial oxygen demand—was significantly greater after heat exposure than after equivalent exercise without heat strain. The same study found that the diastolic pressure-time integral, an index of coronary supply, was significantly lower. The inequality $D > S$ therefore widens on both sides under heat.

More recent positron emission tomography data quantify the demand side directly. When core temperature was raised by 1.5°C, myocardial blood flow increased 2.08-fold in healthy young adults, 1.79-fold in healthy older adults, and 1.64-fold in older adults with coronary artery disease. The absolute increase was similar across groups—approximately 0.6 to 0.8 $\text{mL} \cdot \text{min}^{-1} \cdot \text{g}^{-1}$—but the reserve capacity to meet it differed. In post hoc analysis, imaging evidence of asymptomatic heat-induced myocardial ischemia appeared in 7 of 20 adults with coronary artery disease (35%).

1.2 The Population-Level Consequence

The clinical translation of this inequality is visible in mortality data. A global meta-analysis of temperature and cardiovascular outcomes found that for extreme heat (99th percentile versus the minimum mortality temperature), the pooled relative risk of death from ischemic heart disease was 1.07 (95% CI 1.04–1.10), from stroke 1.10 (95% CI 1.06–1.15), and from heart failure 1.12 (95% CI 1.05–1.19). Each 1°C rise above the reference temperature increased coronary heart disease risk by approximately 2.8%. These are not exotic magnitudes; they are the arithmetic of a demand-supply mismatch applied across millions of exposures.

2 The Renal Mechanism: Flow Redistribution and Cytotoxic Injury

2.1 The Flow Equation

Let Q_{total} be cardiac output and Q_{skin} cutaneous blood flow. The flow available to the renal circulation is, to first approximation,

$$Q_{\text{renal}} = Q_{\text{total}} - Q_{\text{skin}} - Q_{\text{other}}. \tag{3}$$

Heat stress increases Q_{skin} substantially. Classic studies report skin blood flow rising by 6–8 $\text{L} \cdot \text{min}^{-1}$ during severe passive heating with core temperature elevations of 1–2°C. This is not a marginal diversion. It is a reallocation of a large fraction of cardiac output toward the thermoregulatory effector organ, at the expense of the viscera—including the kidneys.

The mechanistic link between this redistribution and renal injury is direct. A 2019 review in the *Journal of Applied Physiology* states that "reductions in renal blood flow are hypothesized to be one of the primary mediators of AKI during heat stress." A separate 2019 review in *Comprehensive Physiology* describes heat-induced reductions in renal and splanchnic blood flow driven by reduced perfusion pressure and sympathetically mediated vasoconstriction. The kidney, in other words, is not passively cooled by the blood it receives; it is actively deprived of that blood so that the skin may be cooled instead.

2.2 The Oxidative Compounding

Ischemia alone would produce reversible injury if brief. The renal lesion becomes more severe because ischemia is accompanied by a second insult: oxidative stress. In exertional heatstroke, rhabdomyolysis releases myoglobin; the free iron from myoglobin catalyzes hydroxyl radical formation, overwhelming tubular antioxidant defenses. A 2026 review describes the resulting injury as a dual mechanism: oxidative ferroptosis from within the tubular cell, plus obstructive cast formation from within the lumen. The mathematical structure of this injury is not a simple threshold but a cumulative dose: repeated subclinical episodes of ischemia-reperfusion, each generating reactive oxygen species, sum over time.

2.3 From Acute to Chronic

The epidemiological consequence is measurable. A meta-analysis cited in the renal literature found an approximately 15% incidence of kidney disease or acute kidney injury among individuals whose occupation requires heat exposure for at least two months per year, six hours per day, five days per week. The hypothesized pathway is repetitive transient injury progressing to chronic tubulointerstitial damage—a pattern consistent with renal biopsies from patients with chronic kidney disease of unknown etiology.

3 The Weaponization Pathway

The mechanisms described above are physiological, not technological. Their weaponization does not require a novel device. It requires the deliberate manipulation of the variables in the equations.

Cardiac weaponization. A population with a high prevalence of coronary artery disease, subjected to sustained heat stress sufficient to raise core temperature by 1.5°C, will experience a predictable incidence of myocardial ischemia—35% in the PET study, with asymptomatic cases invisible until infarction occurs. The weapon is not heat as an agent; it is the selection of a vulnerable population and the withholding of cooling, hydration, and medical care. The mortality risk ratios cited above (1.07–1.12 for extreme heat) become a planning parameter, not a natural hazard.

Renal weaponization. The renal equation is similarly exploitable. Populations dependent on outdoor physical labor—agricultural workers, construction workers, military personnel in training—are already the documented victims of heat-associated kidney injury at 15% incidence.

A deliberate policy of forced labor in heat, without acclimatization or cooling, would convert a recognized occupational hazard into a mechanism of slow attrition. The injury is subclinical for long periods; the damage accumulates before it is detected. This is not speculative in principle: the U.S. torture memos explicitly authorized "exposure to extremes of heat and cold" as an interrogation technique, precisely because thermal stress inflicts severe physiological harm without leaving immediate visible marks.

The weaponization pathway operates through environmental and infrastructural manipulation rather than novel technology. It requires only the deliberate removal of protective conditions—cooling, hydration, rest, medical care—and the selection of physiologically vulnerable populations, such that the resulting harm is individually indistinguishable from natural causes.

4 Conclusion

The heart and kidney are not incidental casualties of heat. They are the organs in which the physics of thermoregulation becomes most legible as pathology. The heart fails when the demand-supply inequality closes; the kidney fails when the flow equation starves it and the oxidative aftermath scars it. These are mathematical relations with parameters—core temperature, cardiac output, renal perfusion, oxidative dose. Parameters can be measured. Measured parameters can be targeted. The weaponization of heat is not the invention of a new ray; it is the exploitation of an old equation.

A note on scope. This paper has modeled the weaponization pathway and stated that weaponization is possible. It has not claimed that any specific state or actor has weaponized heat in the manner described. The physiological mechanisms are documented; the inferential step from mechanism to intent is left to the reader.